A Project Report

On

Microstructural and mechanical analysis of multiwalled CNTs/Mg composite.

Submitted to

Amity University Uttar Pradesh

in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

by

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under the guidance of

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Amity Institute of Aerospace Engineering

AMITY UNIVERSITY UTTAR PRADESH

NOIDA (U.P.)

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Declaration by the student

We, Tolety Eesha and Anmol Gupta, student(s) of B.Tech (AIAE) hereby declare that the project titled "**Microstructural and mechanical analysis of multiwalled CNTs/Mg composite**" which is submitted by us to Department of Amity Institute of Aerospace Engineering, Amity University Uttar Pradesh, Noida, in partial fulfillment of requirement for the award of the degree of Bachelor of Technology in Aerospace Engineering, has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

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To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere

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Abstract

Carbon Nanotubes showing the expectational mechanical properties and enhancement of the other properties have been of much interest in creating some advanced Composites. They exhibit excellent tensile strength and elastic modulus, which is stiffer than steel and three to five times lighter. Products in many fields, composites reinforced CNTs are top of the line. Many concerns about global warming led to the sustainable production of products. In this report, the study of behavior of Mg/CNT surface composite. The composite is fabricated by FSP with little changes for better distribution of CNT in the composite. This fabrication was done on 4 samples. From the 4 samples, 2 samples are the ones in which holes were drilled at distances 20mm from each other and other 2 samples at 10mm distance from each other. After fabrication of these samples, they are investigated for identification of pression of CNT and uniform distribution. From XRF graph where elements of Carbon, Magnesium and Oxygen were, it was confirmed that CNT were implanted during the fabrication. Then using SEM, visual conformation of CNT presence was confirmed. Also, their uniform distribution all through sample could be observed.

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1 Introduction

1.1 Composite materials

Composite materials are synthesised by adding more than two distinguished materials with different physical and chemical properties, which in results gives us a material with enhanced characteristics and properties which were initially not present in the initial ingredients.

The constituent materials are called reinforcement and the matrix. The reinforcement can be in the form of fibres, particles or flakes. While on the other hand, the matrix is generally a polymeric, ceramic, or metallic material.

Composites are desired to be stronger, lighter and more durable than the initial ingredients used which makes the application of composite material in aerospace, automotive, construction and sports equipment vast and dominant.

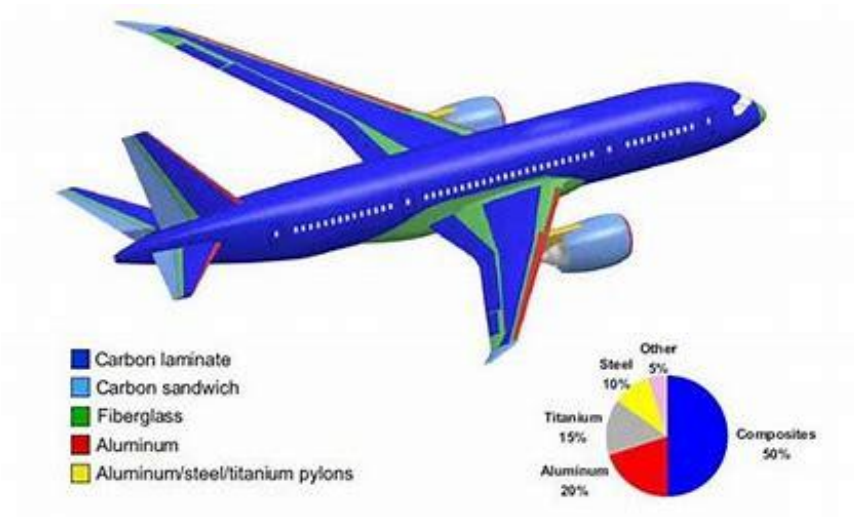

Figure 1 Composites in aerospace sectors

1.1.1 Fibers and Matrices

Fiber is especially deemed to be high-strength and high performance materials which are embedded within the matrix to achieve the goal of having improved strength, stiffness, and other desirable properties. Some of the key materials which are extensively used as fibers are carbon, glass, aramid and are typically characterized by their high aspect ratio and tensile strength properties.

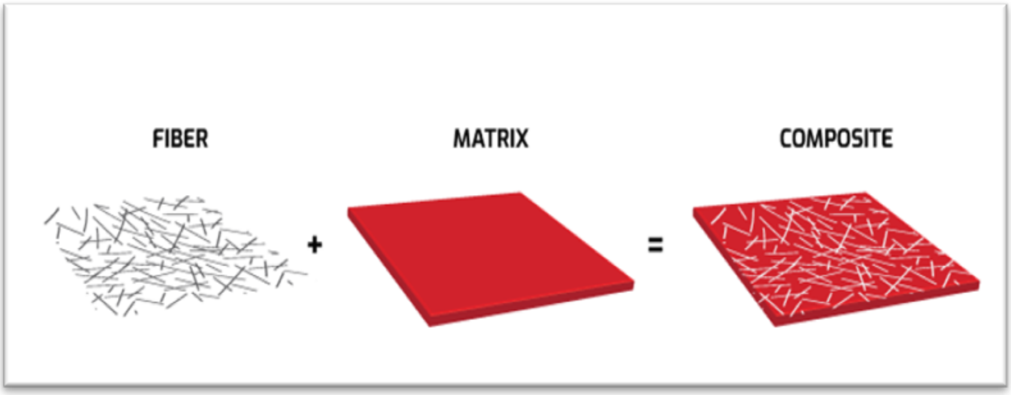

Figure 2 Composite components

On the other hand, matrix materials are supposed to keep the fibers intact and transfer the load between them. It can vary from polymer, ceramic, or metal and are generally chosen on the basis of its ability to exhibit good adhesion with fibers, durability and especially thermal and chemical resistance. The matrix acts as protective covering that supports, resists deformation and also distribute the load evenly across the complete material. The combination of the two discussed ingredients (i.e. fibers and matrices) results in the composite material that has enhanced capabilities than the individual properties.

1.2 Types of Composites

1.2.1 Metal Matrix Composites

Metal matrix composites (MMCs) are advanced materials that contain a metal matrix and one or more other materials. Dispersion materials are usually in the form of granules, fibres or whiskers and are added to the material matrix at 5-50% by volume. The development of MMC was driven by the need to improve the material in applications requiring high strength, hardness, wear resistance, thermal stability, and other properties that cannot be finished with conventional metal or ceramic alone. MMCs provide a unique composition that allows materials to be created for specific applications.

The manufacturing process of MMC can involve a variety of techniques, such as powder metallurgy, casting, and infiltration, all of which produce different microstructures and components in the composite.

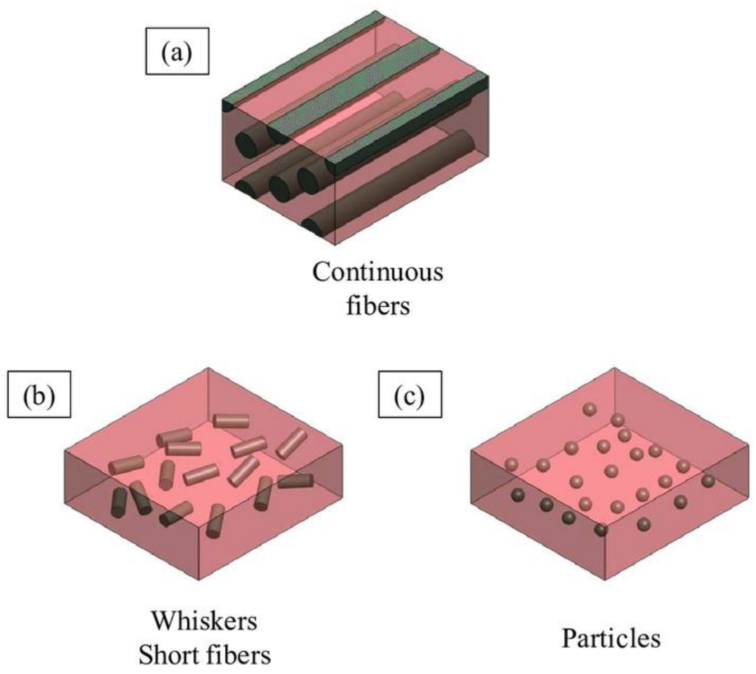

Figure 3 Types of Metal Matrix Composites

In powder metallurgy, the matrix material is mixed with powder dispersed materials, pressed into the desired shape, and then sintered at high temperature. Casting requires that explosives be placed in the mould before pouring the molten matrix. The infiltration will infiltrate the matrix material into the preform of the dispersed material.

MMC can also be produced using spray, which involves spraying molten metal onto a substrate while introducing explosives into the spray. The properties of MMC depend on the composition, processing and microstructure of the composite material. In general, MMC exhibits greater strength, hardness, wear resistance and thermal stability than conventional metals or ceramics. The microstructure of the MMC is important in determining its performance. A uniform distribution of materials dispersed in the matrix is necessary to ensure that the composite exhibits the desired properties. The size, shape and orientation of the particles can affect the composition. For example, the fibers together can increase the strength and stiffness of the composite in certain directions. Applications of MMC are diverse and can be found in many industries, including aerospace, automotive and electronics. In aviation, MMC is used in engine components that require power, robustness and thermal stability, such as turbine blades. In the automotive industry, MMCs are used in brake discs that require wear resistance and thermal stability. In electronics, MMC is used in heatsinks that require high thermal conductivity and stability

1.2.1.1 Magnesium Matrix Composites

Magnesium matrix composites (MMCs) are a class of advanced materials that contain a magnesium alloy matrix reinforced with strong fibres or components. The magnesium matrix provides a lightweight and corrosion resistant base material, while providing mechanical properties such as added strength, durability, hardness and toughness. The application of Magnesium as a matrix material is becoming more and more popular due to its low strength, good strength and good properties. However, the low ductility and poor wear of magnesium alloys limit their use in certain applications. Adding additional materials such as ceramic or metal fibers, awns or particles can improve the properties of magnesium alloys, making them suitable for many applications.

In powder metallurgy, additional materials are mixed with magnesium alloy powder and pressed into the desired shape. In extra virgin casting, reinforcement is placed in the mold and molten magnesium alloy is poured into the mold to allow the reinforcement to leak. The properties of MMC depend on the composition, processing and microstructure of the composite material. MMCs can be found strong, tough and tough, and their properties can be customized for specific applications by selecting magnesium alloys and reinforcements as well as the manufacturing process.

Applications of MMC are very diverse and can be found in many industries such as aerospace, automotive and biomedical. In aviation, MMCs are used in aircraft structures, engine components and satellite structures that require high strength, rigidity and lightness. In the automotive industry, MMC is used in products that require strength and corrosion resistance, such as engine blocks, brake calipers and suspension. In biomedicine, MMCs are used in dental implants and prosthetics that require biocompatibility and good mechanical properties.

1.2.2 Ceramic Matrix Composite (CMC)

Ceramic matrix composites (CMC) are advanced materials that contain ceramic materials and additional materials, usually fibres or materials, in the matrix. The reinforcement provides strength, stiffness and toughness to the composite, while the ceramic matrix provides high temperature stability and corrosion resistance.

CMC has been developed with the best properties for monolithic ceramics with increased strength, toughness and thermal shock resistance. They are also lighter and more corrosion resistant than conventional metal alloys, making them attractive for a wide variety of high temperature and stress applications.

CMCs have been studied for years with a focus on their design, function and use. The most common production methods for CMC are chemical vapour infiltration, melt infiltration and polymer infiltration, and pyrolysis.

Chemical Vapour Sealing involves placing a ceramic material over a porous material made of additional materials such as carbon or silicon carbide fibres. Melt seepage requires molten ceramic material to seep into preforms made of reinforcing materials. Polymer infiltration and pyrolysis involves the impregnation of preforms made of additives with polymer precursors, which are then converted into ceramic materials by heat treatment. The properties of CMC depend on the composition, processing and microstructure of the composite material.

In general, CMCs show higher strength, hardness and thermal stability compared to monolithic ceramics. Adding fibers or materials to the ceramic matrix can increase its hardness, while using different ceramic materials can provide different temperature stability and corrosion resistance. The microstructure of the CMC is important in determining its performance. To ensure that the composite exhibits the desired properties, the reinforcement must be properly dispersed within the matrix. The direction, ratio and volume ratio of the reinforcement also affect the properties of the composite.

Figure 4 CMC application in Turbines

Applications for CMCs are diverse and can be found in industries such as aerospace, defence, and energy and automotive. In aviation, CMC is used in products that require strength, rigidity and thermal stability, such as turbine blades and heat shields. In the energy sector, CMC is used in components such as fuel cells and heat exchangers that require high temperature stability and corrosion resistance. In the automotive industry, CMC is used in products that require high strength, rigidity and thermal stability, such as brake rotors and motors.

1.2.3 Polymer Matrix Composites (PMC)

Polymer matrix composite (PMC) is an advanced material consisting of a polymer matrix reinforced with high-strength fibres or particles. While the polymer matrix acts as a material, it provides added strength, mechanical properties such as strength, hardness and toughness. The polymer material used in PMC may be thermoset or thermoplastic, and the reinforcement material may be organic or inorganic. Polymers used in PMCs include epoxies, polyesters, and plastic esters, while additional materials include carbon, glass, and aramid fibres.

The manufacturing process for PMC is complex and involves several steps, including the manufacture of artificial materials, polymer matrix preparation, and composite fabrication. The most common manufacturing processes for PMCs include filament winding, and compression moulding. In Filament winding, the fibres are wound in a special pattern on a rotating mandrel and a polymer matrix is applied to the fibres using a resin infusion process. In compression moulding, the fibres are laid in a special pattern and placed in a mould, after which a polymer matrix is used for the fibres under heat and pressure.

The properties of PMC depend on the composition, processing and microstructure of the mix. In sports, PMC is used in equipment that requires strength and stiffness, such as tennis rackets and golf club shafts.

1.3 Nano composite materials

A nanocomposite is a material that contains a matrix material (such as a polymer) and tiny particles of discrete nanoscale particles. The word "nano" refers to the size of particles, usually less than 100 nanometres in diameter. Nanomaterial components can be nanoparticles, nanotubes or other nanoscale structures.

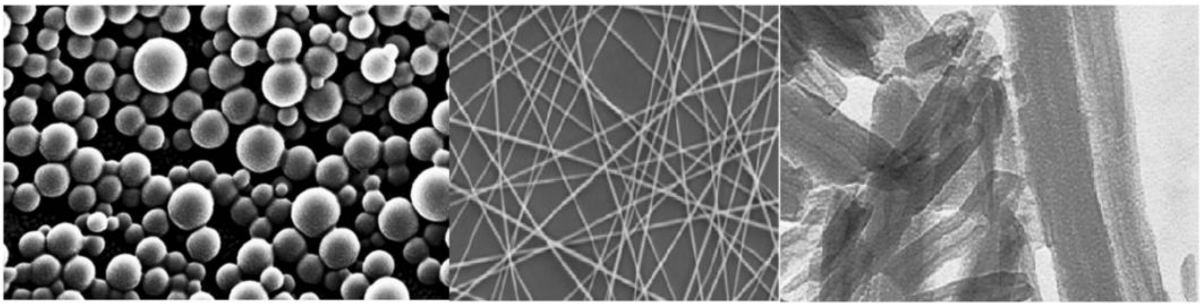

Figure 5 Nanoparticle, Nanofibers, Nano clays

The nanoparticles used in nanocomposites can be made from a variety of materials, including metals, metal oxides, carbon-based materials, and polymers. These materials are often chosen for their physical and chemical properties, such as their high strength, electrical conductivity or optical properties.

Adding nanoparticles to matrix materials can improve the properties of the resulting nanocomposites. For example, the addition of nanoparticles can increase the strength, stiffness and thermal stability of polymers, making them suitable for many high-performance applications. Another advantage of nanocomposites is that they can be designed to exhibit certain properties such as electrical conductivity, magnetism or transparency. This is because the properties of nanoscale particles can be tuned by adjusting their size, shape, and composition. Nanocomposites are also being investigated for biomedical applications.

For example, polymer matrix nanocomposites are designed for drug delivery and tissue engineering scaffolds. Nanoparticles can be used to encapsulate drugs that target tissues or cells throughout the body. Nanocomposites can also be used to create beautiful tissue-forming scaffolds that can be used to repair damaged or diseased tissue.

Despite the potential of nanocomposites, their use poses many challenges. One of the main challenges is to ensure that the nanoparticles are evenly dispersed throughout the matrix. If the nanoparticles are not close to each other, the product can be evaluated as negative and the desired product cannot be found.

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1.3.1 Types of Nano Composite material

There are different types of Nano composites including clay-based Nano composites, metal matrix Nano composites, and polymer-matrix Nano composites.

1.3.1.1 Clay-based Nano Composites

Clay-based nanocomposites are a class of nanocomposites that use clay nanoparticles as a support. They are made from a variety of materials, including clays such as montmorillonite, bentonite, and kaolinite, and synthetic clays such as layered double hydroxides (LDHs) and metal-organic frameworks (MOFs).

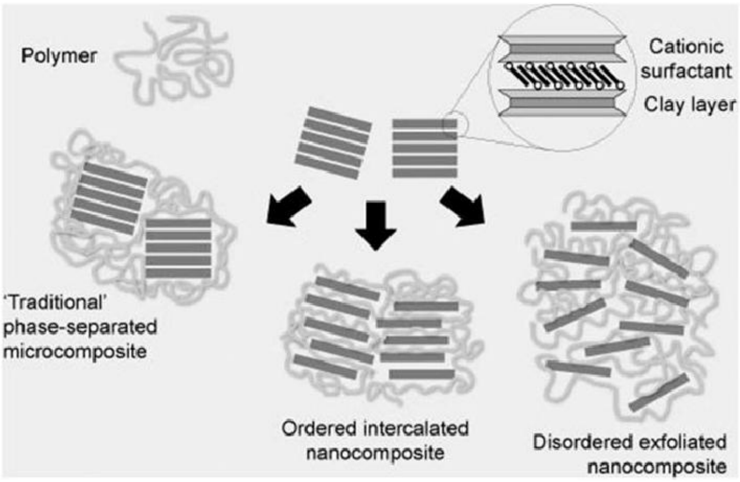


Figure 6 Clay-based Nnano composites

In clay-based nanocomposites, the clay nanoparticles are dispersed in a polymer matrix such as polypropylene, polyethylene or polystyrene. Nanoparticles are often treated with surfactants to improve their dispersion in the matrix and prevent aggregation. An important advantage of clay-based nanocomposites is that they have better physical, thermal and thermal properties than polymer materials. This is because clay nanoparticles have a high aspect ratio, which means they have a larger surface area for that ratio.

Therefore, they interact more strongly with the polymer matrix, which increases the strength and stiffness of the product. In addition, clay nanoparticles can improve the product's barrier properties by increasing resistance to gases and liquids throughout the product.

Clay-based nanocomposites have many applications, including packaging, automotive components and biomedical devices. In the packaging industry, clay-based nanocomposites are used to create films and coatings with enhanced properties that help extend the shelf life of food and other products.

1.3.1.2 Metal-matrix Nano Composites

Metal matrix nanocomposites are composites in which nanoparticles are added to the metal matrix. The nanoparticles used for support are usually smaller than 100 nanometers and can be made of various materials such as carbon nanotubes, graphene, and ceramic particles such as alumina and silicon dioxide.

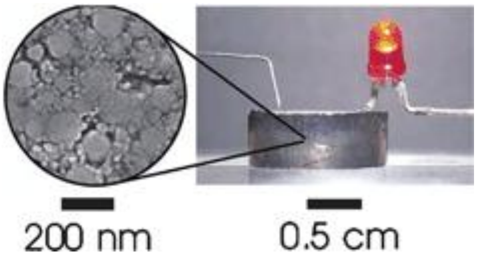


Figure 7 Bismuth Metal Nano composite

In metal matrix nanocomposites, the nanoparticles are dispersed in a metal matrix that can be made of various metals such as aluminium, magnesium, titanium, and copper. Nanoparticles are often added to metal matrices through a process called powder metallurgy, which involves mixing metal powder and nanoparticles together, then compacting and sintering the mixture to form the product. This is because nanoparticles have a different mass-to-volume ratio, which allows them to interact with the metal matrix.

In metal matrix nanocomposites, the nanoparticles are dispersed in a metal matrix that can be made of various metals such as aluminium, magnesium, titanium, and copper. Nanoparticles are often added to metal matrices through a process called powder metallurgy, which involves mixing metal powder and nanoparticles together, then compacting and sintering the mixture to form the product. This is because nanoparticles have a different mass-to-volume ratio, which allows them to interact with the metal matrix.

In the aerospace industry, metal matrix nanocomposites are designed for lightweight, high-strength components that increase fuel efficiency and reduce emissions. In the automotive industry, metal matrix nanocomposites have been designed for brake discs, engine parts and components to improve performance and reduce weight. In the electronics industry, metal matrix nanocomposites have been developed for use in heat sinks, electronic devices, and other electronic devices that require high thermal conductivity.

1.3.1.3 Polymer-matrix Nano Composites

A polymer matrix nanocomposite is a composite material containing polymer materials supported by nanoparticles. They are made from a variety of materials such as clay, carbon nanotubes, graphene, metal oxides, and Nano cellulose.

Polymer matrix nanocomposites are dispersed in a polymer matrix that can be made from thermoplastics such as polyethylene, polypropylene, and polypropylene styrene, as well as various polymers such as nanoparticles, epoxies, and phenolic.

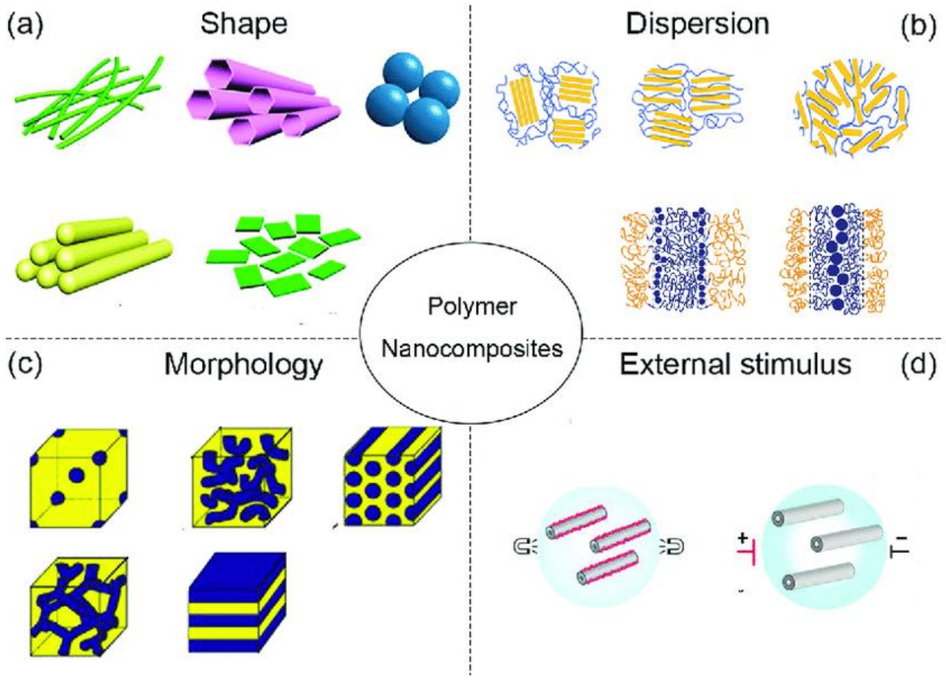


Figure 8 Polymer-matrix Nano Composites

Nanoparticles are usually added to the polymer matrix through a process called melt blending, which involves mixing the polymer and nanoparticles together and then heating the mixture to melt the polymer and melt the nanoparticles. An important advantage of polymer-based nanocomposites is that they exhibit better mechanical, thermal and thermal properties compared to polymer materials.

This is because the nanoparticles have a high aspect ratio, which makes them interact more strongly with the polymer matrix. Therefore, nanocomposites can be stronger, tougher and tougher than pure polymers, with improved thermal stability and barrier properties. In addition, nanoparticles can change the electrical properties of the polymer matrix, making it stronger or improving its dielectric properties. Polymer-based nanocomposites have many applications, including the packaging, automotive and biomedical industries. In the packaging industry, polymer-based nanocomposites are used to create films and coatings with enhanced properties that help extend the shelf life of food and other products.

In the automotive industry, polymer-based nanocomposites are used to create lightweight, fuel-efficient vehicles that increase fuel efficiency and reduce emissions. In the biomedical field, polymer matrix nanocomposites are designed for drug delivery, tissue engineering scaffolds and promising therapies.

1.4 Carbon Nanotubes (CNTs)

Carbon nanotubes (CNTs) are cylindrical structures made of carbon atoms, usually a few nanometres in diameter and a few millimetres long. CNTs have unique electrical, electronic and electrical properties that make them attractive in many applications such as electronic and biomedical engineering and energy storage.

In 1991, Sumio Iijima, who was the first to discover carbon nanotubes, showed that many carbon nanotubes (MWNTs) could be synthesized using arc-evaporated graphite electrodes in a helium atmosphere. Since then, many methods have been developed to produce carbon nanotubes, including chemical vapor deposition (CVD), laser ablation, and high-pressure carbon monoxide (Hi- PCO) methods. The special properties of carbon nanotubes are due to their unique structure, which gives them high aspect ratio, high rigidity, and high thermal and electrical conductivity. Young's carbon nanotube modulus is about 1 TPa, more than 100 times that of steel, and its tensile strength is as high as 130 GPa, several times that of steel. Also, carbon nanotubes have good thermal conductivity, SWNT about 3500 W/mK, MWNT about 300-600 W/mK, and electrical conductivity can be as high as SWNT 1000 S/cm and MWNT 100-300 S/cm. . The high proportions and mechanical strength of carbon Nano Tube make it a good support for composites. When combined with matrix materials such as carbon nanotubes, polymers or metals, they can improve composite properties such as stiffness, strength and toughness. CNTs can also impart other useful properties to composites, such as electrical conductivity, thermal conductivity, and self-lubricating behaviour.

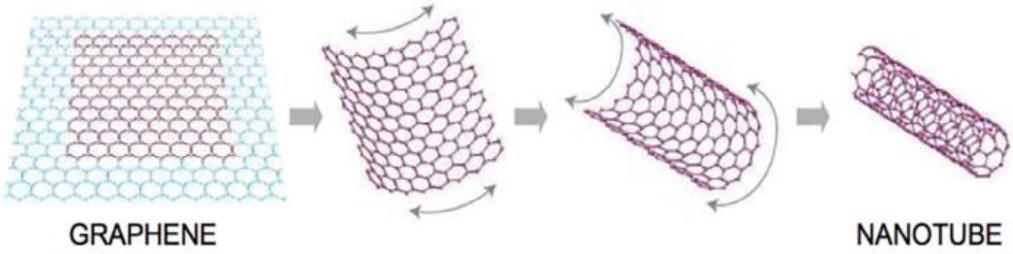


Figure 9 Formation of CNT

Despite their potential applications, there are many challenges to overcome in the synthesis, processing and characterization of CNTs. One of the main challenges is to achieve a high degree of control over the size, shape and properties of CNTs during synthesis. Another challenge is to obtain different carbon nanotubes in the matrix material required to achieve the desired properties in the composite.

1.4.1 Types of Carbon Nanotubes

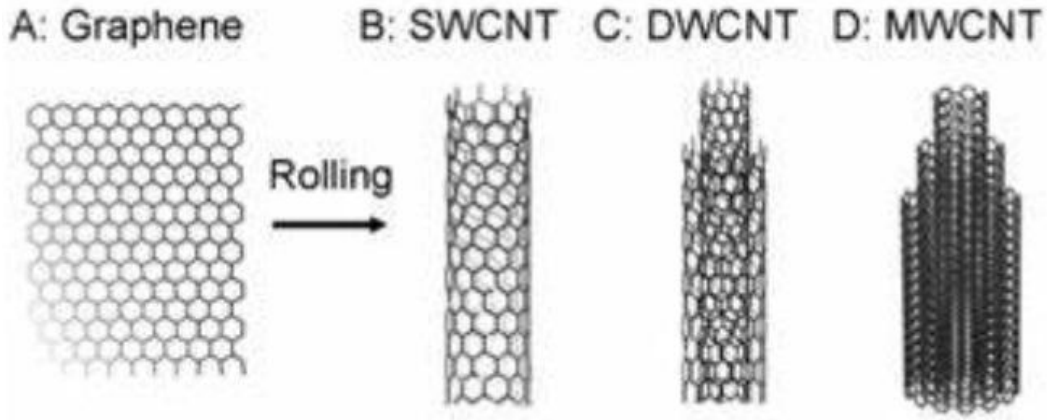


Figure 10 Types of CNT

1.4.1.1 Single-Walled Carbon Nanotubes (SWCNTs)

Single-walled carbon nanotubes (SWCNTs) are one-dimensional cylindrical nanostructures made from a sheet of graphene, a two-dimensional material composed of carbon atoms arranged in a hexagonal lattice. Single-walled carbon nanotubes are usually between 0.4 and 2 nanometres in diameter and can be several micrometres in length. They are known for their unique electrical, mechanical and thermal properties, leading to product searches in many fields. One of the most important properties of single-walled carbon nanotubes is their high tensile strength, which is about 100 times stronger than steel of the same weight.

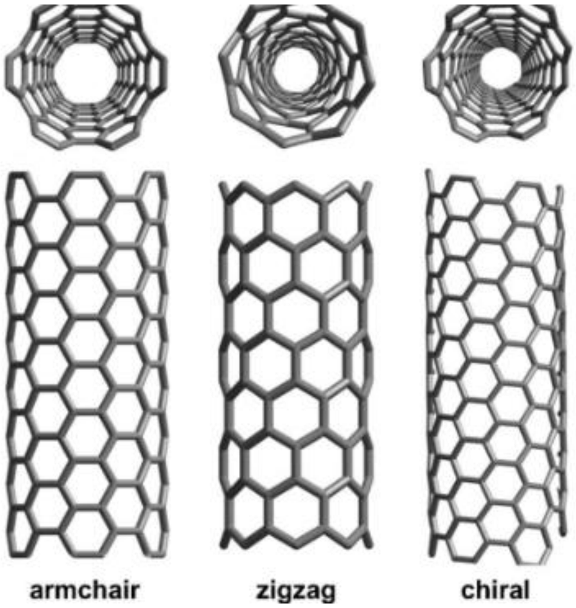


Figure 11 Different types of SWCNTs

This makes them ideal candidates for composites, which can be used to increase the strength and stiffness of materials such as polymers and metals. In addition to its mechanical properties, single-walled carbon nanotubes also have excellent electrical properties, up to 1000 times that of copper. This makes them useful in electronics and other high-performance applications.

SWCNTs also have high thermal conductivity, making them useful for heat dissipation applications. SWCNTs are generally produced using arc discharge, laser cutting or chemical vapour deposition (CVD) methods.

In an arc discharge, a high voltage is applied to a graphite electrode in an inert gas environment, creating a plasma that vaporizes carbon atoms and forms single-walled carbon nanotubes. Laser ablation involves irradiating a graphite target with a laser in a high-temperature chamber, producing single-walled carbon nanotubes by evaporation and condensation. CVD involves the decomposition of hydrocarbon gases over a metal catalyst to produce single-walled carbon nanotubes.

SWNTs are frequently used in research and development due to their unique properties, but SWNTs also have many applications in many fields. For example, they are used as additional electronic components in lithium-ion batteries to improve their performance, and as sensors to detect gases and other substances.

1.4.1.2 Multi-Walled Carbon Nanotubes (MWCNTs)

The full name of MWCNT is Multi-Walled Carbon Nanotube, a type of carbon nanotube with many concentric graphene cylindrical layers inside. MWCNTs are typically 2-100 nanometres in diameter and can have tens to hundreds of layers, each about 0.34 nanometres thick.

MWCNTs can be produced using a variety of methods, including chemical vapor deposition (CVD), arc discharge, laser ablation, and high carbon monoxide (Hi-PCO) processes. Among these methods, CVD is the most widely used method for the synthesis of MWCNTs.

Due to their unique structure, MWCNTs have physical, chemical and mechanical properties such as high tensile strength, stiffness and thermal conductivity. These properties make MWCNTs promising candidates for a variety of applications in fields such as energy storage, electronics, sensors and composite materials. In terms of application, multi-walled carbon nanotubes are widely used as additional materials in composite materials to improve their properties such as strength, hardness and toughness. Because of their high surface area and electrical conductivity, they are also used as electrical components in electronic devices such as supercapacitors and lithiumion batteries.

MWCNTs are also used in Nano electronics, where they are used as building blocks for the fabrication of Nano electronic devices such as transistors, sensors, and memory devices. The unique electrical properties of MWCNTs, such as high carrier density, make them attractive candidates for electrical applications. In addition to the applications mentioned above, MWCNTs have also been explored for biomedical applications such as drug delivery, testing, and tissue engineering. MWCNTs have been shown to be biocompatible and can be functionalized with specific target ligands for cancer cells. Despite the promise of MWCNTs, their commercialization is still hampered by many challenges, including high production costs, difficulties in controlling the synthesis process, and concerns about their potential toxicity. However, research is ongoing to address these issues to fully understand the potential of MWCNTs in various applications.

1.4.1.3 Benefits of MWCNTs

Multi-walled carbon nanotubes (MWCNTs) have many advantages and applications due to their unique properties. Some of the main benefits of MWCNTs are:

- High Strength and Hardness: MWCNTs have high tensile strength and stiffness, making them useful as composite materials. This improves the strength of the material, such as strength, durability and resistance to deformation.

Property	Unit	SWCNTs	MWCNTs
ρ	$g\,m^{-3}$	~ 1.3	~ 1.75
d_t	nm	0.6–0.8	5–50
AR	–	100–10,000	100–10,000
k	$W\,(mK)^{-1}$	3000–6000	3000–6000
ϱ	$\Omega\,m$	$1 \times 10^{-3} - 1 \times 10^{-4}$	$2 \times 10^{-3} - 1 \times 10^{-4}$
σ_{max}	GPa	50–500	10–600
$\bar{E}$	GPa	1500	1000

Table 1 Comparsion of SWCNT and MWCNT

- High electrical and thermal conductivity: MWCNTs are excellent electrical and thermal conductors, making them useful in electronic devices such as sensors, transistors and batteries. MWCNTs can also be used as thermal management materials to dissipate heat from electronic devices.
- Biocompatible: MWCNTs have been shown to be biocompatible, meaning they are harmless to bacteria and tissues. This makes them useful for biomedical applications such as drug delivery or tissue engineering.
- Chemical stability: Multi-walled carbon nanotubes are very resistant to chemical degradation and therefore can be used in harsh environments such as chemical or chemical sensors.
- Wide aspect ratio: MWCNTs have a high aspect ratio, which means they are tall and thin. This could make them useful in applications requiring high surface area such as catalysts or energy storage devices.

1.5 Material selection-Magnesium

Magnesium is a lightweight metal with high strength and a high strength-to-weight ratio, making it a very attractive material in the aerospace industry. The aerospace industry needs materials that are strong, lightweight and corrosion resistant, and magnesium has all of these properties. One of the main benefits of using magnesium in aviation applications is its weight, which helps reduce fuel consumption and increase aircraft range. Magnesium is about 30 percent lighter than aluminium and 75 percent lighter than steel, making it ideal for aircraft components such as airframes, engines and gears.

Magnesium also has an excellent strength-to-weight ratio, meaning it's strong enough to withstand weight and stress without adding weight.

This makes it an ideal material for critical aerospace components such as turbine blades, engine mounts and gearboxes.

In addition, magnesium has good corrosion resistance, which is important for aerospace applications exposed to harsh environments such as high altitudes and high temperatures. Magnesium resists corrosion from salt water, acids and other corrosive substances, making it useful for aircraft components exposed to these conditions.

In general, the use of magnesium in aerospace applications helps to reduce weight, increase fuel efficiency and improve the overall performance of the aircraft. However, the use of magnesium is not without problems, its flammability and workability and communication are difficult.

1.6 Friction Stir Processing (FSP)

The friction stirring process (FSP) is a mechanical process used to change the microstructure and properties of metals and alloys. The process is based on the Friction Stir Welding (FSW) principle, which uses a rotating tool to heat and plasticize the material, allowing it to mix and stir to form bond state words.

In FSP, use the same conversion tool to change the microstructure and material of the file without creating a contract. The tool has a cylindrical pin with a toothed or toothed shoulder, which is inserted into the product to be processed, generating heat and plasticizing the product. The device is then transported to the site where the materials are mixed and combined to form microstructures. The FSP process can be used to improve the properties of grain size resulting in strength and ductility. It can also be used as a good, stable secondary material to improve wear, corrosion and other mechanical properties. The technique can be used to create layers with tenable microstructures, allowing components of selected materials to be modified in specific areas. The FSP process has many advantages over other commercial methods. It is a solid state process that does not involve melting or solidification, eliminating problems such as porosity, cracking and deformation.

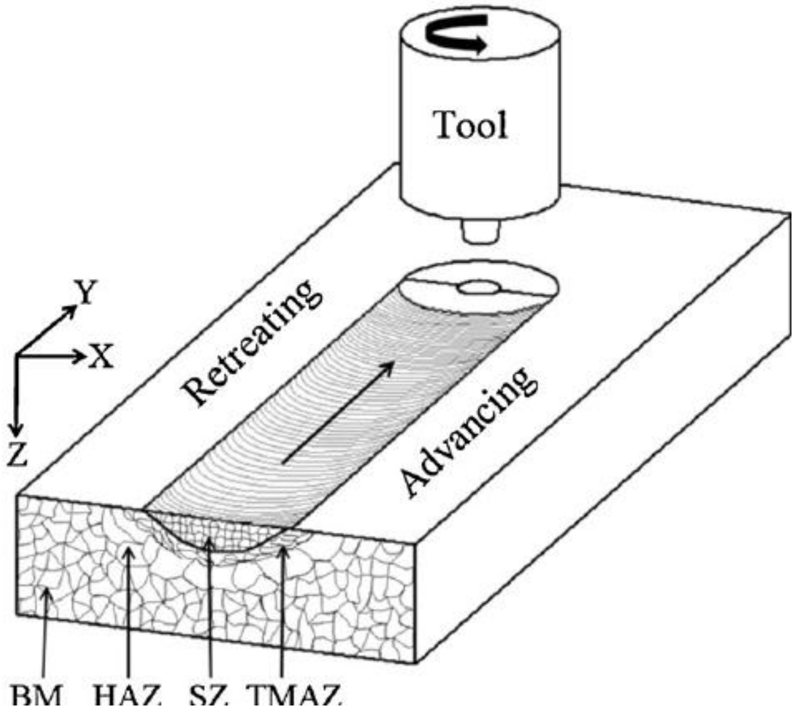


Figure 12 Friction Stir Processing (FSP)

In conclusion, friction mixing is a promising steady-state processing technique that can be used to modify the microstructure and properties of metals and alloys. Compared to other devices, this system has many advantages and is widely used in many places. Further research in this area could lead to new developments and create new technologies with greater potential for commercial use.

1.7 SEM Analysis

SEM (Scanning Electron Microscopy) is a powerful analytical technique for studying the surface morphology and structure of various materials. It provides high resolution images and measurements of samples, making it an essential tool in information science, biology and many other fields.

The SEM works by focusing a high-energy beam onto the sample surface. Electrons interact with atoms in the structure to produce various types of matter, including secondary electrons, back electrons, and X-rays. These signals are detected and processed to create a sample image. The second electron signal provides information about the topography of the sample surface. When an incoming electron hits a sample, it can strike a second electron from the surface of the sample detected by the SEM detector to form an image. On the other hand, the backscattered electron signal provides information about the composition and crystal structure of the sample. Backscattered electrons are produced when an incident beam of electrons interacts with the atomic nuclei in a sample. The number of backscattered electrons returned to the detector by depends on the element's atomic number, so the backscattered electrons can be used to map the distribution in the sample.

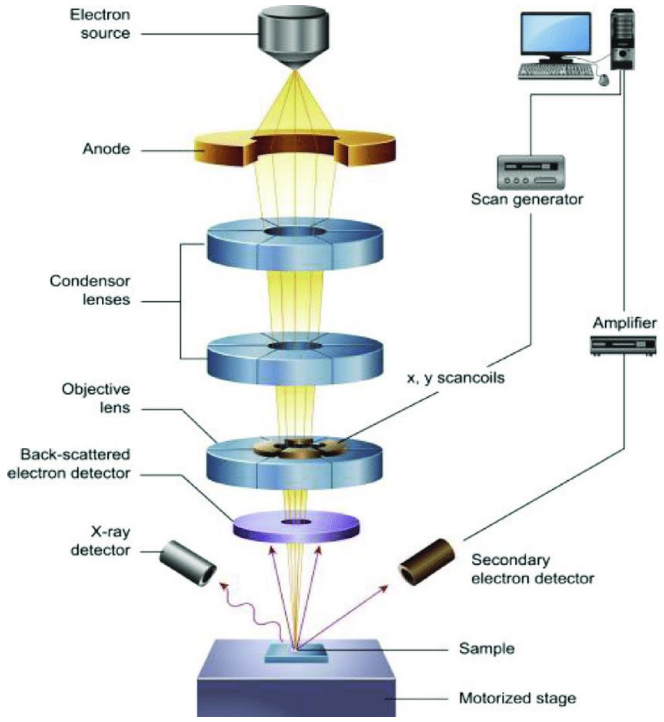


Figure 13 Working of SEM

SEM images can be produced in a variety of ways, including secondary electron microscopy, retrograde electron microscopy, and X-ray imaging. These images can be viewed in real time or captured as digital images for further analysis and processing. In addition to images, SEM can also be used for quantitative measurements of samples. For example, the method can be used to measure the size and distribution of particles in the sample, the thickness of the thin film, and the roughness of the surface. SEM can also be used for microanalysis, which provides information about the details of the structure of a particular part of the surface.

Overall, SEM is a versatile and powerful technique that provides high-resolution images and quantitative measurements of data. Its ability to provide detailed information about the morphology, composition and structure of surface materials makes it an important tool in many research.

1.8 XRF Analysis

The analytical method known as X-ray fluorescence, or XRF, is frequently employed to figure out the elemental makeup of materials. The idea behind it is that when a substance is subjected to high-energy X-rays, its atoms may absorb the energy and release distinctive fluorescent X-rays.

A sample is exposed to X-ray radiation during an XRF study, which is generally produced using an X-ray tube. The excited atoms in the sample release inner-shell electrons as a result of the high-energy X-rays. Typical X-rays are released when the electrons migrate to fill the resultant vacancies. Each of the X-rays that are released has a distinct energy that enables their identification.

A detector, such as a scintillation detector or a solid-state detector, is used to detect and analyse the produced X-rays. The elemental makeup of the sample may be ascertained by measuring the energy and intensity of the X-rays using the detector.

XRF analysis may be carried out on a variety of sample types, including solids, liquids, and powders, and is non-destructive. It is often employed in a number of disciplines, including as geology, environmental science, archaeology, materials science, and quality control in sectors like mining, manufacturing, and pharmaceuticals.

Wavelength-dispersive X-ray fluorescence (WD-XRF) and energy-dispersive X-ray fluorescence (ED-XRF) are the two primary forms of XRF analysis. With the help of crystals, WD-XRF disperses X-rays according to specific wavelengths, making it possible to measure X-ray energy precisely. The detector used in ED-XRF detects the X-ray energy directly, obviating the necessity for crystal dispersion. The choice of either approach depends on the particular specifications of the analysis, and both offer benefits.

2 Experimental Procedure

2.1 Raw Materials

There are two main raw materials used to produce the Mg/CNT which are Pure Magnesium and Multi-walled CNT. Magnesium of 99.8% purity was obtained from an online vendor of Nextgen based in Mumbai. It was obtained in five pieces of 10mm x10mmX 100mm. Multi-walled CNT was used from the minor project, where the CNT was produced from sugarcane waste. The resulting CNT was used in this project, they have inner diameter of 10-20nm and outer diameter to be 30-50nm and length of 10 microns with surface area of 250-300 m^2 /g.

2 

Figure 14 Raw Materials: Magnesium and CNT

2.2 Tool Selection

There were two tools in the CNC machine for the experimental procedure. They are:

- Milling tool: This tool is used for making holes or to cut through the material. Tool was obtained from market, with 4 mm fruit diameter and 2mm cutting diameter. It is made from high steel speed.
- FSP tool: This tool was made on campus in amity university Noida. The material used was Tool steel . It was made using shaper machine with parts which are pin, shoulder, fixture. Dimensions are

➢ Pin: Diameter- 3mm, length -1.5 mm

➢ Shoulder: Diameter- 9mm, length -20 mm

➢ Fixture: Diameter- 3mm, length – 10mm

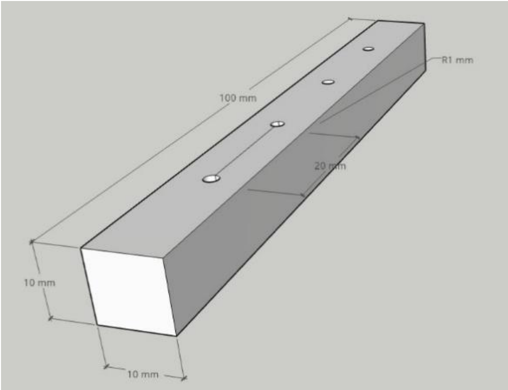
Figure 15 tools used: Milling tool and FSP tool

2.3 Experiment

For the Production and Process uses CNC machine. The CNC machine set-up used for the process is from Amity University, Mechanical Department. The model of the CNC machine is CNC MILLING-HEM12.Magnesium pieces were mounted on a fixture in CNC milling Machine and milling tool was mounted on tool fixture.

Figure 16 Experiment set-up: CNC Milling machine

Before the holes were drilled, the pieces were polished and smoothed out. This polish process might be a small step but it is very important. If the piece is not polished or smoothed properly, the FSP process with not be as desired and arise a lot of issues. In two pieces, holes were drilled at distance of 10mm and on the two samples 20mm.



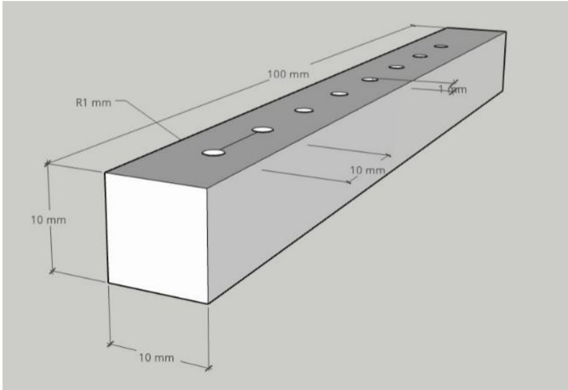


Figure 17 Diagram of holes at a distance of 10 mm

Figure 18 Diagram of holes at a distacne of 20mm

Figure 19 Experimental representation of holes in piece

CNT was filled in the holes with help of mini spatula. Then, the tool fixture is changes into FSP tool. The tool was then programmed to go through the material. The constant tool rotated at rate of 900 rpm with a travel speed of 5 mm/min. There was no tool tilt. The effect of the experiment is on surface level, the samples are cut in half using the CNC milling and 1 mm milling tool. Now the dimension of the samples is 4mm x 10mm x 100mm.

Figure 20 FSP Process

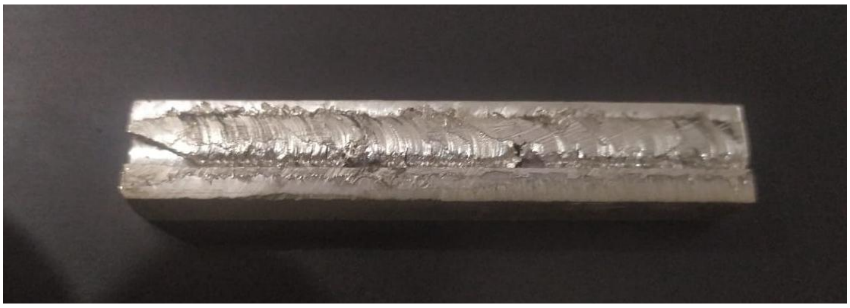


Figure 21 Mg/CNT sample

2.4 Testing

For testing, the main objective is to observe the sample for the identification and distribution of the MWCNT in Mg pieces. The indication of the MWCNT is done by the XRF and the identification is done by the SEM. When the XRF was performed with 4 no. of iterations for efficient results. In SEM sample test were preformed at 20.00kv ETH which represents the electron beam is fired onto the sample, an energy of 20.00 kV is used to accelerate the electrons. The Working Distance on the sample is 6.5nm and Magnification is from 50.00K X to 10.00 K X during SEM analysis.

Figure 22 SEM and XRF set-up

3 Results and Discussion

The Mg/CNT composite were produced by Friction stir Processing. XRF is conducted for two samples, one having holes at a distance of 10 mm and the other one having 20 mm. The graph that is obtained from two graphs represent the presence of Carbon, Magnesium and Oxygen at their intensities and energy levels

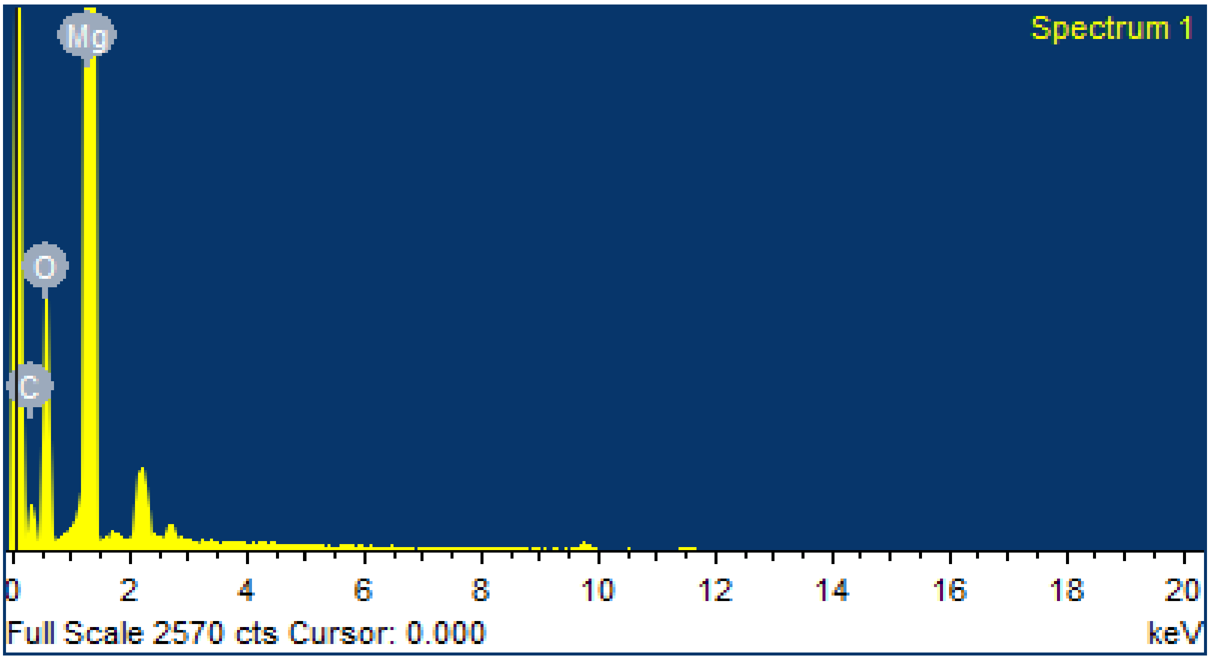


Figure 23 XRF graph

.Both the samples produced same graphs but the data about the Weight percentages changed. The data is represented on Table 2 for holes at a distance of 10mm and Table 3 for holes at a distance of 20mm.

Element	Weight%
C	9.63
O	28.21
Mg	62.16
Total	100.00

Table 2 XRF data (10mm)

Element	Weight%
C	16.44
O	24.77
Mg	58.79
Total	100.00

Table 3 XRF data (20mm)

From the observation of the table, the weight percentage of the Carbon in the Mg is 9.63% for holes at a distance of 10mm and 16.44% for holes at a distance of 20 mm. Carbon here represents CNT here as CNT is made of Carbon only. Also, observation are done that Oxygen is found and most probably composition it is found is MgO(Magnesium oxide). Compound MgO is found due to environmental interaction. Some studies shows that Mg interaction with environment over times cause MgO. Also there are studies showing that MgO in the composite has not effected the strength rather has increased in some cases. From comparing to many productions process for Mg/CNT the percentage of MWCNT present is higher.

SEM analysis is done next to represent the distribution of MWCNT and also to show that they are embedded in Mg. Figures from 24 to 26 show SEM images of the samples. In the Figure 24 MWCNT can be seen MWCNT embedded into the Mg. In this image, there is clearly tube like structure which are CNT, where half of them is embedded into Mg piece.

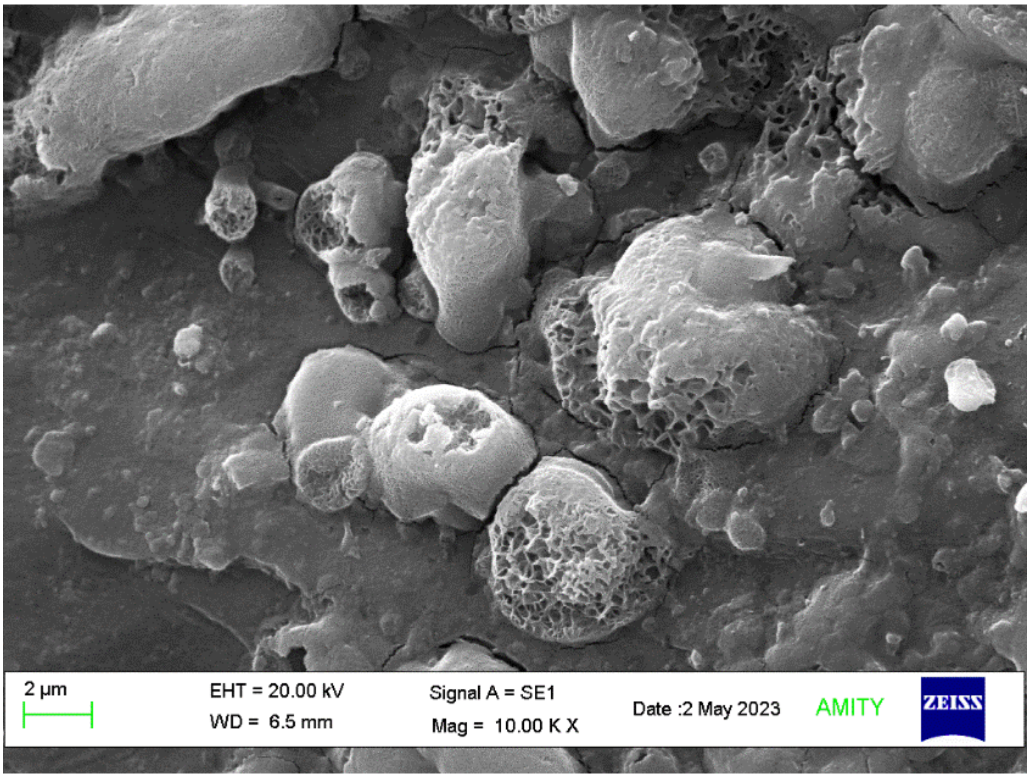


Figure 24 SEM Image-1 of Sample

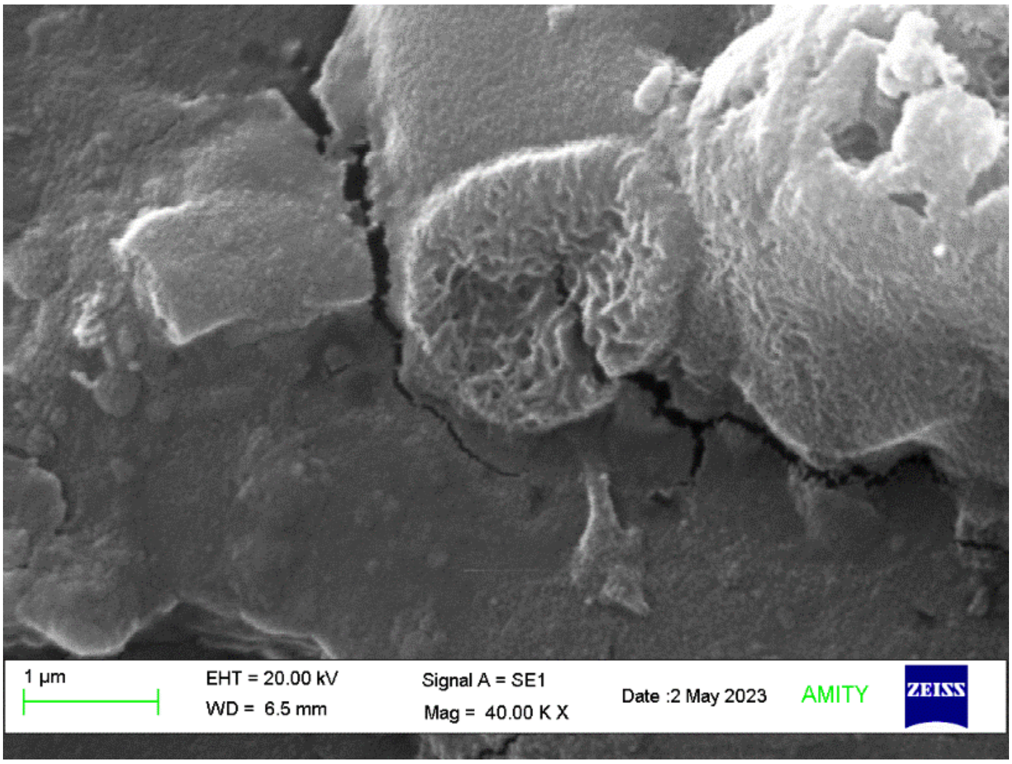


Figure 25 SEM Image-2

In Figure 25, shows a magnified image of MWCNT is implanted in Mg. Figure 26, is SEM imagining at same magnification at different areas showing that the MWCNT is distributed thought out the sample uniformly.

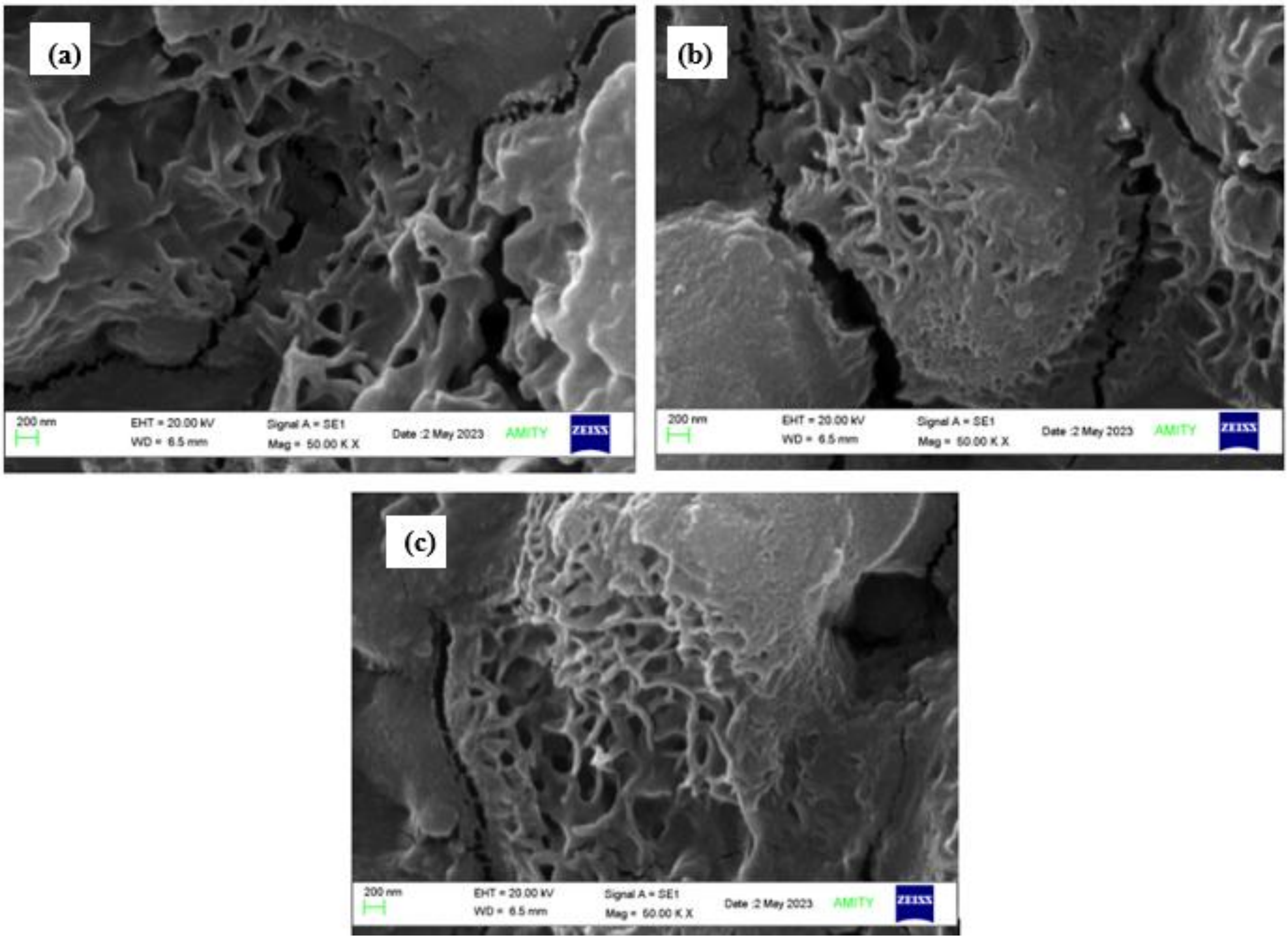


Figure 26 SEM images at different areas in the sample

This process of making hole has optimised the old FSP process of making a groove in making an improvement in the surface to make surface composite. As material was put in holes and tool was sent in very low travel speed. This low speed has helped in making the material soft due to friction and let the CNT present in the holes get implanted into Mg. Also, the CNT in holes has travelled with Tool for a distance, because of which there is distribution though-out the sample.

4 Conclusion

The Mg/CNT Surface composite was successfully produced by new and optimised FSP. The microstructure of the composite is investigated by XRF and SEM imaging. The results obtained are:

- MWCNTs are dispersed into the Mg of 99.8 purity using the FSP and Optimised method.
- The low travel speed of the tool aided in proper embedding of the MWCNT in Mg pieces.
- From XRF, it could be confirmed that the sample consisted of CNT by the amount of Carbon. Also, we could observe that samples where the holes were made with 20mm have more Carbon content than 10 mm.
- SEM imaging could confirm that MWCNT were in fact embedded in Mg along the length of the sample. Also, we could see the uniform distribution obtained by this FSP process.

5 References

1) Synthesis of bimetallic NiMo/MgO catalyst for catalytic conversion of waste plastics (polypropylene) to carbon nanotubes (CNTs) via chemical vapour deposition method - H.U. Modekwe , M. Mamo , K. Moothi , M.O. Daramola
2) Sugarcane-converted Graphene-Like Material for the Super-High Adsorption of Organic Pollutants from Water via Co-assembly Mechanisms - Xin Xiao, Baoliang Chen, Lizhong Zhu, and Jerald L. Schnoor.
3) Carbon nanotube reinforced composites: Potential and current challenges Amal M.K. Esawi , Mahmoud M. Farag
4) Potential and prospective implementation of carbon nanotubes on next generation aircraft and space vehicles: A review of current and expected applications in aerospace sciences -Omid Gohardani , Maialen Chapartegui Elola , Cristina Elizetxea
5) Carbon Nanotubes By Chris Scoville, Robin Cole, Jason Hogg, Omar Farooque, and Archie Russell
6) Characterization of Nanomaterials Produced from Sugarcane Bagasse Joner Oliveira Alves, Jorge Alberto Soares Tenório, Chuanwei Zhuo, Yiannis Angelo Levendis.
7) Understanding and control of interactions between carbon nanotubes and polymers for manufacturing of high-performance composite materials - Cécile A.C. Chazota, , A. John Hart
8) Carbon Nanotube Film Based Multifunctional Composite Materials: An Overview- D. W. Lee1 and J. W. Seol
9) Carbon Nanotubes--the Route Toward Applications- Ray H. Baughman et al.
10) Carbon nanotube (CNT) reinforced magnesium matrix composites: The effect of CNT ratio on their mechanical and corrosive properties- Yakup Say, Omer Guler, Burak Dikici
11) Chickpea peel waste as sustainable precursor for synthesis of fuorescent carbon nanotubes for bioimaging application- Vikram Singh, Shovon Chatterjee, Mahendra Palecha, Pratik Sen, Bushra Ateeq ,Vivek Verma.
12) Potential release scenarios for carbon nanotubes used in composites- Bernd Nowack , Raymond M. David , Heinz Fissan , Howard Morris , Jo Anne Shatkin , Michael Stintz , Richard Zepp , Derk Brouwer

13) Synthesis of Multi-Walled Carbon Nanotubes from Plastic Waste Using a Stainless-Steel CVD Reactor as Catalyst- Pranav K. Tripathi, ID , Shane Durbach, and Neil J. Coville.
14) Nano-composite multi-wall carbon nanotubes using poly (p-phenylene terephthalamide) for enhanced electric conductivity, Authors: Azza M. Mazrouaa, Nahla A. Mansour, M. Yahia Abed, Mona A. Youssif, M.A Shenashen, Md. Rabiul Awual
15) Koziol K, Shaffer M, Windle A. Three dimensional internal order in multiwalled carbon nanotubes grown by chemical vapour deposition. Adv. mater. 2005;17;7607763. CNTs contain zigzag edges and are crystallized.
16) Iijima S. Helical microtubules of graphitic carbon. Nature 1991;354;56758
17) Bethune DS, Klang CH, DeVries MS, Gorman G, Savoy R, Cobalt catalysed growth of carbon nanotubes with single7atomic7layer walls. Nature 1993;363;6057607.
18) Buchholz DB, Doherty SP, Chang RPH. Mechanism for the growth of multiwalled carbon7nanotubes from carbon black. Carbon 2003;41;162571634
19) Mitchell DR, Brown RM, Spires TL, Romanovicz DK, Lagow RJ. The synthesis of megatubes: new dimensions in carbon materials. Inorg. Chem. 2001;40;275172755.
20) Mitchell DR, Brown RM, Spires TL, Romanovicz DK, Lagow RJ. The synthesis of megatubes: new dimensions in carbon materials. Inorg. Chem. 2001;40;275172755.
21) Ebbesen TW, Ajayan PM. Large7scale synthesis of carbon nanotubes. Nature 1992;358;2207222
22) Novel trends for synthesis of carbon nanostructures from agricultural wastes - Nady A. Fathy , Altaf H. Bastab, Vivian F. Lotfy
23) Carbon nanotubes synthesis using carbonization of pretreated rice straw through chemical vapor deposition of camphor Nady A. Fathy
24) A review featuring the fundamentals and advancements of polymer/CNT nanocomposite application in aerospace industry - Anila Iqbal, Aamer Saeed, Anwar Ul-Hamid
25) T. Kuzumaki, O. Ujiie, H. Ichinose, K. Ito, Adv. Eng. Mater. 2 (7) (2000) 416–418.
26) R.Z. Ma, J. Wu, B.Q. Wei, J. Liang, D.H. Wu, J. Mater. Sci. 33 (1998) 5243–5246.
27) K.T. Lao, D. Hui, Carbon 40 (2002) 1605–1606.
28) P. Calvert, Nature 399 (1999) 210–211.

29) E.T. Thostenson, Z.F. Ren, T.W. Chou, Compos. Sci. Technol. 61 (2001) 1899–1912.
30) M.M.J. Treacy, T.W. Ebbesen, J.M. Gibson, Nature 381 (1996) 678–680. \
31) .W. Wong, P.E. Sheehan, C.M. Lieber, Science 277 (1997) 1971–1975.
32) R.F. Service, Science 281 (1998) 940–942. [9] J.P. Salvetat, J.M. Bonard, H.K. Tomson, A.J. Kulik, L. Forro, W. Benoit, L. Zuppiroli, Appl. Phys. A 69 (1999) 255–260.
33) M.R. Falvo, C.J. Clary, R.M. Taylor, V. Chi, F.P. Brooks, S. Washburn, R. Superfine, Nature 389 (1997) 582–584.
34) S. Frank, P. Poncharal, Z.L. Wang, W.A. de Heer, Science 280 (1998) 1744–1746.
35) A. Batchtold, C. Strunk, J.P. Salvetant, J.M. Bonard, L. Forro, T. Nussbaumer, C. Schonenberger, Nature 397 (1999) 673–675.
36) S.J. Trans, A.R.M. Verschueren, C. Dekker, Nature 393 (1998) 49–52.
37) J. Kong, N.R. Franklin, C.W. Zhou, M.G. Chapline, S. Peng, K.J. Cho, H.J. Dai, Science 287 (2000) 622–625.
38) A. Watabe, A. Oshida, T. Kobayashi, H. Toda, J. Jpn. Inst. Light Met. 49 (1999) 149–154.
39) H.J. Liu, H. Fujii, K. Nogi, Mater. Sci. Technol. 20 (2004) 399–402.
40) K. Ohishi, T.R. Mcnelley, Metall. Trans. A 35A (2004) 2951–2961.
41) J.Q. Su, T.W. Nelson, C.J. Sterling, Scr. Mater. 52 (2005) 135–140.
42) D.C. Hofmann, K.S. Vecchio, Mater. Sci. Eng. A 402 (2005) 234–241.
43) R.S. Mishra, Z.Y. Ma, Mater. Sci. Eng. R 50 (2005) 1–78.
44) R.S. Mishra, Z.Y. Ma, I. Charit, Mater. Sci. Eng. A 341 (2003) 307– 310.
45) K.V. Java, K.K. Sankaran, J.J. Rushau, Metall. Mater. Trans. A 31A (2000) 2181–2188.
46) Y.S. Sato, H. Kokawa, Metall. Trans. A 32A (2001) 3023–3031.
47) S.H.C. Park, Y.S. Sato, H. Kokawa, Scr. Mater. 49 (2003) 161–166.
48) Z.Y. Ma, R.S. Mishra, Scr. Mater. 53 (2005) 75–80
49)] K. Kondoh, T. Threrujirapapong, H. Imai, J. Umeda, B. Fugetsu, Compos. Sci. Technol. 69 (2009) 1077–1081
50) K. Kondoh, H. Fukuda, H. Imai, B. Fugetsu, Magnesium Technology, TMS, 2008, pp. 289–291.
51) S.J. Splinter, N.S. McIntyre, W.N. Lennard, K. Griffiths, G. Palumbo, Surf. Sci. 292 (1993) 130–144.